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ARTICLE

Shifting Representations: Understanding How CEO Cognitive Flexibility Fosters Digital Product Innovation in Incumbent Firms

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Abstract

While digital product innovation offers unprecedented opportunities for incumbent firms to create competitive advantages, its fluid and iterative nature presents distinctive cognitive challenges. There is limited empirical literature explaining the role of constructive cognitive characteristics of executives in differentiating the successful pursuit of digital product innovation. Drawing on upper echelons theory and the attention-based view, we examine how chief executive officer (CEO) cognitive flexibility enables incumbent firms to pursue digital product innovation. Through a mixed-methods approach combining a field survey of 178 machine-building firms and a scenario-based experiment with 134 participants, we demonstrate that CEOs with higher cognitive flexibility achieve superior digital product innovation outcomes by facilitating insightful information acquisition and processing. This relationship is strengthened when CEOs engage in more boundary spanning activities and when firms possess greater social capital. Our study contributes to strategic leadership research by demonstrating how CEO cognitive flexibility enables incumbent firms to navigate the cognitive demands of digital product innovation, while enriching our understanding of how adaptive attention patterns shape strategic adaptation in increasingly digitized environments.

摘要

尽管数字产品创新为在位企业创造竞争优势提供了前所未有的机遇， 但其流动性和迭代性特征带来了独特的认知挑战。然而， 现有实证文献对高管的建设性认知特征在数字产品创新成功追求中的差异化作用解释不足。本研究基于高阶梯队理论和注意力基础观(ABV)，考察了首席执行官(CEO)的认知灵活性如何使在位企业能够推进数字产品创新。通过混合研究方法, 结合对178家机械制造企业的实地调查和134名参与者的情景实验，研究表明，具有较高认知灵活性的CEO能够通过促进富有洞察力的信息获取和处理， 实现更优的数字产品创新成果。当CEO从事更多的边界跨越活动以及企业拥有更强的社会资本时， 这种关系得到加强。本研究通过揭示CEO认知灵活性如何使在位企业应对数字产品创新的认知需求, 对战略领导力研究做出了贡献, 同时丰富了我们对适应性注意力模式如何塑造日益数字化环境中战略适应的理解。

Keywords: attention-based view; CEO; cognitive flexibility; digital product innovation; upper echelons

关键词: CEO; 认知灵活性; 高阶梯队; 注意力基础观; 数字产品创新

Introduction

Innovation is the lifeblood of organizations; it is essential for firms to grow, succeed, and maintain competitive advantages (Raffaelli, Glynn, & Tushman, 2019). Digital technologies such as artificial intelligence, cloud computing, virtual reality, blockchain, and the Internet of Things are rapidly penetrating businesses, significantly transforming the nature of product innovation

(Nambisan, Lyytinen, Majchrzak, & Song, 2017; Yoo et al., 2024; Yoo, Henfridsson, & Lyytinen, 2010). This transformation has led to growing academic and practical interest in digital product innovation – novel products or services that are embodied in or enabled by digital technologies (Firk, Gehrke, Hanelt, & Wolff, 2022; Lyytinen, Yoo, & Boland, 2016; Pesch, Endres, & Bouncken, 2021).

Digital product innovation exhibits fluid and iterative nature that creates both opportunities and challenges for firms. At its core, digital products are 'malleable, editable, and open' (Nambisan et al., 2017: 225), continuing to evolve even after market release through continuous updates and iterations (Lehmann & Recker, 2022; Yoo, Boland, Lyytinen, & Majchrzak, 2012). While this enables firms to create unbounded design flexibility across the product lifecycle (Wang, Henfridsson, Nandhakumar, & Yoo, 2022), it also demands a significant shift in how products are conceptualized from discrete, completed entities to fluid, continuously evolving offerings that blur traditional boundaries between development and deployment (Firk et al., 2022; Henfridsson & Yoo, 2014). This shift is particularly challenging for incumbent firms in traditional industrial contexts, where established product development processes are built around fixed release cycles and stable product boundaries (Röth, Schweitzer, & Spieth, 2023; Svahn, Mathiassen, & Lindgren, 2017). Even incumbent firms that show initial ambition in digital product innovation typically 'end up with incremental optimizations of existing operations and processes' rather than embracing its fluid, iterative nature (Moschko, Blazevic, & Piller, 2023: 507).

The distinctive characteristics of digital product innovation create unique cognitive demands for incumbent firms' executives. Unlike traditional product development that follows predictable patterns, digital product innovation challenges their established cognitive representations by requiring rapid shifts in attention: from completed to continuously evolving offerings, from product to ecosystem thinking, and from predetermined to emergent value creation (Nambisan et al., 2017). Yet they typically rely on fixed cognitive representations deeply ingrained in historical manufacturing contexts, leading to struggles in dealing with these unique demands. For example, Volberda, Khanagha, Baden-Fuller, Mihalache, and Birkinshaw (2021) found that successful digital product innovation requires incumbent executives to continuously shift their cognitive frames to match the fluid and iterative nature of digital products, which is a capability they often lack. Given these cognitive challenges inherent in the transition to digital product innovation, a salient question is: *what role does chief executive officer (CEO) cognition play in enabling incumbent firms to successfully pursue digital product innovation?*

While empirical research has examined various organizational structures, external relationships, and strategic orientations (e.g., Bockelmann, Werder, Recker, Lehmann, & Bendig, 2024; Ceipek, Hautz, De Massis, Matzler, & Ardito, 2021; Liu, Dong, Ying, & Jiao, 2021), the cognitive characteristics that enable executives to process and act upon the distinctive demands of digital product innovation have been largely overlooked. This gap is particularly significant given that conceptual and case-based studies have revealed how digital product innovation creates unique demands that challenge executives' established cognitive representations (e.g., Bunduchi, Crisan-Mitra, Salanta, & Crisan, 2022; McCarthy, O'Raghallaigh, Kelleher, & Adam, 2025). To shed further light on this topic, we examine how cognitive flexibility, defined as the capability to rapidly adjust cognitive representations and attention based on changing situations (Kiss, Libaers, Barr, Wang, & Zachary, 2020; Martin & Rubin, 1995), enables executives, particularly CEOs, to meet the distinctive demands of digital product innovation. Drawing on upper echelons theory and the attention-based view (ABV), we propose that cognitive flexibility is crucial for incumbent firms' digital product innovation success. Specifically, cognitively flexible CEOs can dynamically shift attentional focus across multiple strategic domains and direct attention toward novel scenarios when responding to emerging opportunities while facilitating the identification and integration of novel insights that drive innovative digital solutions (Kiss et al., 2020).

This study further proposes a conceptual framework based on the ABV to explore the contingency of the relationship between CEO cognitive flexibility and digital product innovation. As a

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scarce resource, how attention is allocated to particular issues and answers has important implications for firm activities and performance. The ABV emphasizes that attention structures within organizations are crucial determinants of how decision-makers notice, encode, and process information from their environment (Ocasio, 1997, 2011). Specifically, the theory suggests that the effectiveness of attention allocation depends on two key organizational mechanisms: the 'communication channels and procedures' that regulate information flow, and the 'social relationships that regulate and control the distribution and allocation of issues and answers' (Ocasio, 1997: 188). These mechanisms not only shape what information reaches decision-makers but also influence how they interpret and act upon that information. Following this theoretical logic, we explore two contextual factors that embody these distinct attention-structuring mechanisms: CEO boundary spanning (Marrone, 2010; Tushman & Scanlan, 1981), which shapes the communication channels through which CEOs access and process external information, and firm social capital (Arregle, Hitt, Sirmon, & Very, 2007), which represents the networks that influence how information and resources are distributed within and across organizational boundaries.

To test our hypotheses, we employed a mixed-methods research design combining a field survey (Study 1) and a scenario-based experiment (Study 2). In Study 1, we conducted a survey of 178 machine-building firms in China to examine the relationship between CEO cognitive flexibility and digital product innovation, along with its boundary conditions. Using both perceptual measures and objective patent data, we found that CEO cognitive flexibility was positively associated with digital product innovation. This relationship was strengthened when CEOs engaged in more boundary spanning activities and when firms possessed greater social capital. To further investigate the underlying psychological mechanisms, Study 2 employed a controlled experiment with 134 participants in a simulated decision-making context. The results demonstrated that individuals with high cognitive flexibility were more inclined to pursue digital product innovation compared to those with low cognitive flexibility and this relationship was mediated by insightful information acquisition.

Our research makes contributions to the literature in several significant ways. First, we advance understanding of the cognitive microfoundations of organizational adaptation that enable incumbent firms to successfully pursue digital product innovation (Bunduchi et al., 2022; Hadjielias, Dada, Cruz, Zekas, Christofi, & Sakka, 2021; McCarthy et al., 2025). While previous research has primarily focused on organizational-level factors (Bockelmann et al., 2024; Ceipek et al., 2021), we illuminate how CEO cognitive flexibility enables firms to adapt the fluid, iterative nature of digital product innovation. Second, we extend the ABV by elucidating how cognitive flexibility shapes attention allocation in complex digital environments, demonstrating the dynamic interplay between individual cognitive capabilities and structural elements in determining innovation outcomes (Brielmaier & Friesl, 2023; Ocasio, Laamanen, & Vaara, 2018). Specifically, we show how cognitive flexibility works through communication channels (manifested in boundary spanning) and social relationships (embodied in firm social capital) to enable effective attention allocation. Third, prior research on strategic leadership has emphasized the importance of understanding cognitive underpinnings of organizational adaptation (Eggers & Kaplan, 2009; Tripsas & Gavetti, 2000). We advance this literature by theoretically elaborating and empirically validating cognitive flexibility as a crucial yet underexplored cognitive characteristic that enables executives to navigate increasingly fluid digitized environments.

Theoretical Background

Digital Product Innovation in Incumbent Firms

Digital technologies are revolutionizing traditional product domains, significantly transforming the nature of innovation (Verganti, Vendraminelli, & Iansiti, 2020; Yoo et al., 2024). It is particularly challenging for incumbent firms, which navigate the transition from established product development paradigms to the fluid, iterative nature of digital innovation processes (Nambisan et al., 2017; Svahn et al., 2017). While digital product innovation offers unprecedented opportunities for value creation and competitive advantages, its characteristics create significant challenges for incumbent firms, influencing not only their ability to generate digital innovations but also to successfully commercialize these innovations in the market (Oberländer, Röglinger, & Rosemann, 2021; Vial, 2019). Unlike traditional product development that operates within clear temporal phases and spatial boundaries, digital products exist in a state of perpetual evolution (Pesch et al., 2021; Yoo et al., 2012). They continue to evolve post-release through continuous updates and iterations, with development and market deployment occurring simultaneously (Lehmann & Recker, 2022). This 'ever-in-the-making' nature conflicts with incumbent firms' structured development cycles and established quality control processes.

The hybrid nature of digital products creates new possibilities for functionality and value creation, as physical components provide the foundational capabilities while digital elements enable feature expansion (Wang, 2021; Yoo et al., 2010). However, for incumbent firms, it requires developing new digital capabilities while maintaining excellence in manufacturing, which is difficult to achieve (Moschko et al., 2023; Stanko & Rindfleisch, 2023). The combinatorial potential changes the nature of innovation strategy, requiring to think beyond predetermined feature sets to consider open-ended possibilities for value creation (Bharadwaj, El Sawy, Pavlou, & Venkatraman, 2013; Nambisan et al., 2017). Moreover, digital products create distinctive forms of value creation through data-driven network effects and ecosystem dynamics (Nambisan, Zahra, & Luo, 2019; Wang, 2021). Unlike traditional products whose value proposition is largely self-contained, digital products often become more valuable as they collect and analyze more user data and connect with more complementary products and services (Lyytinen et al., 2016; Verganti et al., 2020). This networked value creation generates complex interdependencies between products, users, and complementors, which acquires incumbent firms to transition from linear value chains to ecosystem orchestration.

These potential challenges of digital product innovation have motivated scholars to examine its stimulus. In Table 1, we include papers that examine antecedents of digital product innovation. The table reveals three key streams of research: studies examining organizational structures and processes (e.g., formalization, alliance portfolios), external relationships and orientation (e.g., market orientation, boundary spanning), and qualitative investigations of cognitive aspects. While the first two streams have provided valuable insights into organizational enablers of digital product innovation, the cognitive dimension, particularly how executives process and act upon the distinctive demands of digital product innovation, remains undertheorized in quantitative empirical research. The established mental models in traditional product development become increasingly inadequate when confronting digital products' fluid and iterative nature. Executives need to flexibly understand contradictory paradigms: fixed versus fluid boundaries, product versus ecosystem thinking, predetermined versus emergent value creation. Hence, this study specifically examines cognitive flexibility as a potential mechanism for addressing these cognitive demands.

Cognitive Flexibility

Cognitive flexibility reflects the adaptive capability that helps humans pursue complex tasks by enabling them to rapidly shift attention patterns, manage contradictory demands, and find novel, adaptable solutions to changing situations (Dajani & Uddin, 2015; Ionescu, 2012; Spiro & Jehng, 1990). Prior research on cognitive flexibility has highlighted three crucial elements: the awareness of available alternatives in specific situations, the willingness to be flexible and adapt to these situations, and the self-efficacy to implement adaptive responses (Dennis & Vander Wal, 2010; Martin & Rubin, 1995). It enables individuals to flexibly allocate attention across different information domains and engage with novel or complex scenarios that demand different cognitive approaches (Liu, Xu, Yu, Wu, & Wang, 2024). While cognitive flexibility's importance for strategic decision-making is well-established, its operationalization presents both empirical and conceptual challenges (Laureiro-Martínez & Brusoni, 2018). Self-reported measures such as the Cognitive Flexibility Scale (Martin &

Table 1. Summary of articles on antecedents of digital product innovation

Paper	Method	Sample	Dependent variable(s)	Independent variable(s)	Findings
Pesch, Endres, and Bouncken (2021)	Survey	395 manufacturing firms	Digital production innovation performance	Formalization	Driving digital product innovation performance and digital patent growth benefit from formalized procedures and structures.
Röth, Schweitzer, and Spieth (2023)	Survey	129 machine-building firms	Digital new market creation	Formalization	Incumbents relying on formalization and decision agility can introducing novel digital innovations and new value to their customers.
Zhao, Peng, Iqbal, and Wan (2023)	Survey	223 Chinese high-tech firms	Digital innovation	Market orientation	Both proactive and reactive market orientation positively correlates with digital innovation.
Shi, Cui, and Kurnia (2023)	Survey	209 Chinese high-tech firms	Digital innovation quality	Boundary object, boundary spanners	Project coordinators can directly influence digital innovation quality, whereas interorganizational systems tools cannot.
Ceipek, Hautz, De Massis, Matzler, and Ardito (2021)	Panel data	46 publicly traded German firms	Internet of Things exploration index	Family management	Lower investments in IoT innovations of an exploratory nature as family involvement in management increases.
Hanelt, Firk, Hilebrandt, and Kolbe (2021)	Panel data	23 automotive manufacturers	Digital patents	Digital mergers & acquisitions	Digital mergers & acquisitions enable firms to drive digital innovation.
Bockelmann, Werder, Recker, Lehmann, and Bendig (2024)	Panel data	550 US firms	Volume and quality of digital patents	Alliance portfolio characteristics	Alliances for digital innovations require a different configuration when compared with alliances for non-digital innovations.
Firk, Gehrke, Hanelt, and Wolff (2022)	Panel data	305 US industrial firms	Digital innovation	TMT digital knowledge	Top management team digital knowledge is positively associated with digital innovation.
Liu, Dong, Ying, and Jiao (2021)	Panel data	600 Chinese manufacturing firms	Digital patents	Organizational Status	High- and low-status manufacturing firms are more likely to be involved in digital innovations than their middle-status counterparts.

(*Continued*)

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Table 1. (*Continued.*)

Paper	Method	Sample	Dependent variable(s)	Independent variable(s)	Findings
Wang, Li, Tian, and Hou (2023)	Mixed-methods	1676 unique Chinese firms	Digital patents	Government digital initiative intensity	Government digital initiative intensity is conducive to stimulating firm managerial digital attention, thereby promoting digital innovation.
McCarthy, O'Raghallaigh, Kelleher, and Adam (2025)	Single case study	A health IoT project	–	–	Actors can choose various socio-cognitive modes of knowledge integration during the initiation stage of digital innovation networks.
Hadjielias et al. (2021)	Multiple case study	3 innovation teams	–	–	Digital innovation team cognition emerges and transforms in teamwork within innovation stages, and intersects with one another in a cyclical and reinforcing manner.
Moschko, Blazevic, and Piller (2023)	Multiple case study	8 manufacturing firms	–	–	Digitizing manufacturing system needs established firms to balance the complexities inherent in digitization and manage conflicting goals.
Bunduchi, Crisan-Mitra, Salanta, and Crisan (2022)	Multiple case study	15 computer software firms	–	–	Founders' cognitive frames about product changes can manage the perpetual incompleteness of digital products.
de Paula et al. (2023)	Multiple case study	45 interviewees with practitioners and educators	–	–	Top managers can act as 'cognizers' to reduce the complexity of digital innovation by fostering shared mental models.
Pershina, Soppe, and Thune (2019)	Multiple case study	9 serious games producers	–	–	Boundary-spanning tools can facilitate the cross-domain collaboration of specialists to jointly create digital innovation.

Rubin, 1995) offer advantages for field research, capturing individuals' dispositional tendencies but also entailing limitations such as social desirability bias. However, performance-based measures such as task-switching paradigms (Rogers & Monsell, 1995) provide complementary insights but may not fully capture the contextual and motivational factors.

The strategic value of cognitive flexibility emerges primarily through its influence on information processing capabilities that underpin strategic decision-making. When individuals demonstrate higher levels of cognitive flexibility, they exhibit enhanced patterns of information processing across multiple cognitive domains (Laureiro-Martínez & Brusoni, 2018). Rather than filtering information through rigid mental models, cognitively flexible executives can more readily recognize and integrate unconventional or novel information that may challenge existing assumptions, enabling them to identify and capitalize on insightful information (Kiss et al., 2020). This enhanced processing manifests in both the breadth and depth of their information engagement: they actively seek out diverse information sources beyond conventional channels, demonstrate greater thoroughness in their analytical processes, and show increased capacity to synthesize novel insights with existing knowledge structures (Furr, Cavarretta, & Garg, 2012; Steinbach, Gamache, & Johnson, 2019). It is not simply about handling more information, but rather about different approaches to discovering, interpreting, and integrating diverse information streams into actionable strategic insights that can drive organizational adaptation and innovation (Dennis & Vander Wal, 2010; Souitaris & Maestro, 2010).

Prior research has indicated that cognitive flexibility becomes particularly valuable when executives encounter strategic contexts characterized by novel information processing demands (Helfat & Martin, 2015). As argued earlier, digital product innovation presents such a context through its distinctiveness (Nambisan et al., 2017). Its characteristics create unique information processing demands that differ from traditional product development paradigms, where executives, particularly CEOs, need to simultaneously process and integrate insights about product architecture, development patterns, and value creation mechanisms (Firk et al., 2022; McCarthy et al., 2025). Therefore, as we detail next, we expect cognitive flexibility to be a critical determinant of digital product innovation by enabling CEOs to better discover insights and act upon the distinctive information patterns inherent in digital products.

Hypotheses Development

Cognitive Flexibility and Digital Product Innovation

We combine upper echelons theory and the ABV to develop reasoning about the relationship between CEO cognitive flexibility and digital product innovation. Upper echelons theory suggests that firms' strategic choices are linked to their executives' filtering and processing of information from external environments (Hambrick, 2007; Hambrick & Mason, 1984). This aligns with the ABV's core premise that organizational outcomes emerge from the bounded rationality in decision-makers' attention to information and environmental signals, balancing 'issues' (e.g., digital disruption) and 'answers' (e.g., innovation strategies) (Gavetti, Greve, Levinthal, & Ocasio, 2012; Ocasio, 1997, 2011). Together, the theories provide a dual lens: while upper echelons theory emphasizes who filters information (executives' cognitive traits), the ABV clarifies how they filter it (attentional allocation).

Attention is a key limitation on information processing in complex digital environments, making it crucial for CEOs to allocate their limited attention rationally and efficiently (Ocasio et al., 2018). Unlike traditional innovation contexts where attention allocation may follow stable routines, digital product innovation demands dynamic attentional agility. Cognitively flexible CEOs demonstrate enhanced capabilities in rapid attention switching without experiencing the typical negative consequences such as cognitive strain or decision errors (Dennis & Vander Wal, 2010; Laureiro-Martínez & Brusoni, 2018). This cognitive adaptability enables concurrent processing of multi-domain information streams spanning operational metrics, emerging technologies, and ecosystem interdependencies (Kiss et al., 2020). It aligns with the core demands of digital product innovation where fluid

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development cycles require continuous monitoring and integration of diverse technology and market inputs (Lyytinen et al., 2016; Nylén & Holmström, 2015).

Additionally, the layered modular architecture of digital products integrates physical components with digital elements, creating a tension between manufacturing logic and generative innovation paradigms (Yoo et al., 2010). While traditional product development adheres to linear, digital components enable iterative reconfiguration and ecosystem-centric value creation. This structural divergence generates a fragmented information landscape where critical insights emerge from reconciling opposing logics, such as fixed production parameters with open-ended digital adaptability. Incumbent firms' CEOs often struggle to address this duality as entrenched mental models prioritize physical constraints over digital possibilities (Volberda et al., 2021). Cognitively flexible CEOs, however, process these contradictions by maintaining openness to both domains. They synthesize insights from manufacturing limitations (e.g., material scalability, supply chain dependencies) with digital potential (e.g., data-driven customization, platform integration), enabling the design of hybrid solutions. This capability to reframe tensions as synergistic opportunities is pivotal, driving the transformation of legacy offerings into digitally augmented innovations.

Finally, the value of cognitive flexibility extends beyond innovation generation to shape how effectively firms commercialize digital product innovations. Digital products exist in a state of perpetual evolution, requiring firms to simultaneously manage innovation development and market deployment (Nambisan et al., 2017). Cognitively flexible CEOs more effectively process and integrate diverse market signals about evolving customer needs and ecosystem dynamics. This enhanced information processing capability enables better alignment between innovations and market demands. Their ability to recognize and act upon insights facilitates experimentation with novel business models and value capture mechanisms, moving beyond traditional product-centric thinking to embrace ecosystem-based revenue models (Bharadwaj et al., 2013). In contrast, cognitively rigid CEOs are more likely to neglect signals conflicting with linear development logics, which impedes recognition of fluid digital possibilities (Laureiro-Martínez & Brusoni, 2018). This mismatch intensifies when rigid cognitions confront the temporal dynamics of digital products, which require continuous deployment cycles. Overall, cognitively rigid CEOs prioritize marginal adjustments to existing offerings while underinvesting in digital assets and capabilities, thereby hindering the pursuit of digital product innovation. Accordingly, we predict:

Hypothesis 1 (H1): CEO cognitive flexibility is positively related to digital product innovation in incumbent firms.

In our main hypothesis, drawing on the ABV, we propose that CEO cognitive flexibility promote incumbent firms' digital product innovation by enabling flexible attentional allocation, which is an adaptive process of shifting focus to identify and act on novel insights (e.g., emerging technologies and user needs). This aligns with ABV's core premise: innovation outcomes emerge from how decision-makers allocate limited attention to specific 'issues' (e.g., digital disruption) and 'answers' (e.g., innovation strategies) within structurally constrained environments.

However, the ABV further emphasizes that attentional focus is not solely an individual trait but is structurally distributed, which is shaped by the firm's communication channels, social relationships, and resource configurations (Joseph, Laureiro-Martinez, Nigam, Ocasio, & Rerup, 2024; Li, Pan, Yang, & Tse, 2022). Thus, while cognitive flexibility determines a CEO's capability to adapt attention, the effectiveness of this adaptation largely depends on contextual factors that regulate information flow and interpretation. We identify two critical attention structures that moderate the cognitive flexibility–digital product innovation relationship: CEO boundary spanning and firm social capital. As a communication channel, boundary spanning determines the diversity and novelty of external information entering the CEO's attentional sphere. As a social relationship embedded in trust and reciprocity, social capital governs how CEOs interpret and act on insights.

Moderating Effects of CEO Boundary Spanning

Boundary spanning encompasses activities through which organizational members establish connections across boundaries to link internal operations with external constituents (Marrone, 2010; Tushman & Scanlan, 1981). It enables CEOs to gather and translate external information, build relationships with diverse stakeholders, and facilitate knowledge transfer across organizational boundaries (Faraj & Yan, 2009). Recent research demonstrates that boundary spanning shapes how firms identify and interpret technological opportunities, access novel knowledge domains, and integrate external insights into their innovation processes (Pershina, Soppe, & Thune, 2019; Shi, Cui, & Kurnia, 2023). Prior research positions boundary spanning activities as crucial attention structures that regulate both the channels through which decision-makers access information and the cognitive frameworks they use to process this information (Dahlander, O'Mahony, & Gann, 2016). Drawing on this theoretical perspective, we examine how CEO boundary spanning moderates the relationship between CEO cognitive flexibility and digital product innovation.

The ABV suggests that executives' attention is inherently limited, requiring them to selectively focus on specific environmental signals while necessarily filtering others (Ocasio, 1997). Cognitive flexibility enables CEOs to dynamically shift their attention across different information domains, but this capability's value depends critically on the richness and diversity of available information streams. Boundary spanning activities create structured channels through which cognitively flexible CEOs can access and process varied technological perspectives, market signals, and innovation practices (Tippmann, Scott, & Parker, 2017). These channels expose CEOs to divergent thinking patterns, novel technological frameworks, and alternative innovation approaches. As boundary spanning intensifies, cognitively flexible CEOs can more effectively leverage their adaptive attention patterns to identify non-obvious connections between physical and digital domains, recognize emerging technological opportunities, and synthesize insights from different stakeholder perspectives (Zhong, Ma, Tong, Zhang, & Xie, 2021).

Boundary spanning further amplifies cognitive flexibility's impact by enriching the contexts in which attention patterns manifest. Extant literature emphasizes that attention structures not only filter information but also shape how that information is interpreted and integrated into decision-making processes (Ocasio et al., 2018; Vuori & Huy, 2016). When cognitively flexible CEOs actively span boundaries, they encounter diverse technological frameworks and innovation logics that expand their interpretive repertoire. This exposure enables them to more effectively reconcile seemingly contradictory demands in digital product innovation, such as balancing fixed manufacturing constraints with fluid digital architectures, or integrating traditional product logic with platform-based value creation (Yoo et al., 2010). Through boundary spanning interactions, cognitively flexible CEOs develop richer mental models that incorporate both established industry knowledge and emerging digital paradigms, enabling more nuanced attention allocation to innovation opportunities (Maula, Keil, & Zahra, 2013; Tippmann et al., 2017).

Without active boundary spanning, cognitive flexibility's impact on digital product innovation may be constrained by limited exposure to diverse attention sources. When boundary spanning is low, cognitively flexible CEOs lack the rich contextual information and diverse perspectives needed to fully leverage their adaptive attention allocation capabilities. Their ability to recognize novel digital opportunities or challenge existing innovation assumptions becomes limited by narrow information exposure, despite their underlying capability for flexible thinking (Joseph & Wilson, 2018). Based on these insights, we posit:

Hypothesis 2 (H2): The positive relationship between CEO cognitive flexibility and digital product innovation in incumbent firms is stronger when the CEO engages in more boundary spanning activities.

Moderating Effects of Firm Social Capital

Social capital represents a firm's accumulated network of relationships that facilitate resource exchange and information flows through established channels of trust and reciprocity (Arregle et al., 2007; Gölgeci & Kuivalainen, 2020). These network structures, developed through long-term interactions and reputation building with diverse stakeholders including suppliers, customers, industry partners, and research institutions, create institutionalized pathways for knowledge sharing, resource mobilization, and collective learning (Chen, Chen, & Huang, 2013; Zahra, 2010). Through the lens of ABV, social capital constitutes a critical attention structure that shapes how firms distribute and control information flows across organizational boundaries (Ocasio, 2011). This network of relationships influences not only the accessibility and quality of strategic information but also the social context through which this information is interpreted and integrated into decision-making processes (White, Hoskisson, Yiu, & Bruton, 2008).

We argue that firm social capital enhances cognitive flexibility's impact on digital product innovation by enriching the social context of attention allocation. The ABV emphasizes that attention patterns are deeply embedded within social relationships that regulate both access to and interpretation of strategic information (Joseph et al., 2024). When firms possess strong social capital, cognitively flexible CEOs can leverage established trust relationships to access deeper insights about technological trajectories, market evolution, and innovation opportunities. These trusted network connections reduce information asymmetries and facilitate more nuanced interpretation of strategic signals, allowing cognitively flexible CEOs to more effectively adjust their mental models in response to emerging digital opportunities (Tippmann et al., 2017). For instance, strong relationships with technology partners can provide early insights into emerging digital capabilities, while trusted customer relationships offer deep understanding of evolving user needs and preferences.

Moreover, social capital provides access to complementary resources that make cognitive flexibility more valuable for digital product innovation. Strong network relationships often come with preferential access to technological capabilities, market intelligence, and innovation partnerships that can support digital initiatives (Arregle et al., 2007; Fey & Birkinshaw, 2005). Cognitively flexible CEOs can leverage these network resources more effectively, rapidly identifying and combining complementary capabilities to support digital product innovation efforts. The combination of cognitive adaptability and resource access through social capital enables firms to more quickly experiment with and implement digital product innovations. Conversely, limited social capital may constrain how effectively cognitive flexibility translates into innovation outcomes. Without strong network relationships, even highly flexible CEOs may struggle to access the rich contextual information needed to validate and refine their strategic insights. Limited social capital also restricts access to complementary resources and capabilities that could support digital innovation initiatives, potentially constraining the value of cognitive flexibility. Thus, we propose:

> *Hypothesis 3 (H3): The positive relationship between CEO cognitive flexibility and digital product innovation in incumbent firms is stronger when the firm has more social capital.*

Overall, we summarize the theoretical framework in Figure 1.

Overview of Studies

We employed a mixed-method research design, combining a field survey (Study 1) and a scenario-based experiment (Study 2) to test our hypotheses. This approach leverages complementary operationalizations of cognitive flexibility, that is, dispositional self-report measures that capture CEOs' behavioral tendencies in organizational settings, and performance-based tasks that assess cognitive switching ability under controlled conditions. Specifically, Study 1 examines the relationship between

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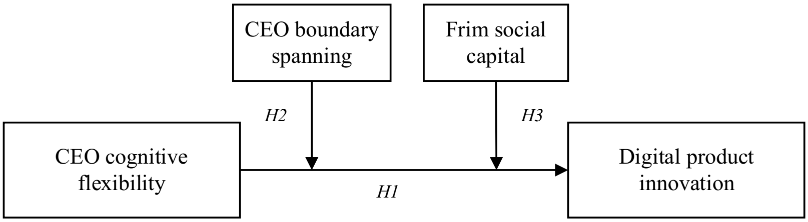

Figure 1. Theoretical framework

CEO cognitive flexibility and digital product innovation through a field survey, along with its boundary conditions. It allows us to capture how cognitively flexible CEOs navigate innovation decisions within their organizational contexts. Study 2 employs a controlled experiment to establish causality and unpack the underlying psychological mechanisms. By manipulating cognitive flexibility via a switching task and simulating digital product innovation decisions, we demonstrate that the relationship operates through insightful information acquisition. The convergence of findings across different measures strengthens the robustness of our theoretical model, demonstrating that cognitive flexibility facilitates digital product innovation whether assessed as a cognitive disposition or ability.

Study 1: Field Survey

Sample and Data Collection

Our empirical setting focuses on machine-building firms in China, a choice that offers several compelling advantages for studying the relationship between CEO cognitive flexibility and digital product innovation. First, as per the 'Digital China Development Report (2023)' released by the Cyberspace Administration of China, the size of China's digital economy surged to RMB 50.2 trillion in 2022, contributing 41.5 percent to the country's GDP. Digitization has become an inextricable theme for both the current and future development of Chinese firms, which provides a rich sample pool related to firms' digital practices. Recent research has conducted empirical investigations into the relationship between CEO cognitive characteristics and Chinese firms' innovation strategies or performance in the digitization context (Zhang, Lu, & Wang, 2024).

Second, while digital technologies have catalyzed systemic changes across corporate strategies, business models, and organizational structures, machine-building firms' primary avenue for digitization lies in embedding digital technologies within product innovation processes (Röth et al., 2023). This creates a focused context where the tensions between traditional manufacturing practices and digital innovation are especially pronounced. Moreover, despite major opportunities, digital product innovation practices remain non-pervasive in machine-building firms due to significant uncertainties and organizational inertia (Liu et al., 2021). This variation is crucial for our research as it allows us to examine how CEO cognitive flexibility influences firms' ability to overcome these barriers and successfully pursue digital product innovation.

To test our hypotheses, we conducted a survey and random sampling techniques were used to collect data from Jiangsu, Zhejiang, and Liaoning, where there are more developed manufacturing industries. The survey was originally created in English and then translated into Chinese employing the back-translation process to ensure conceptual equivalence. Prior to the formal survey, to guarantee the clarity, readability, and completeness of questionnaire items, we randomly selected 10 CEOs from machine-building firms to conduct a pre-test (not included in the sample). Some revisions and refinements were made to the questionnaire according to their feedback. Then, a total of 800 machine-building firms were randomly selected from the directory of provincial industrial associations. We

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distributed formal questionnaires through both on-site and online surveys. An invitation for participation was sent by email and telephone to the CEO of each sampled firm, and of which 396 accepted the invitation.

To conduct the formal investigation, we collaborated with a professional research company to identify trained interviewers conducting face-to-face interviews with CEOs, and an online platform administering questionnaires electronically. This approach is recommended for obtaining reliable and high-quality survey information in China (Jean, Kim, Zhou, & Cavusgil, 2021; Zhao, Peng, Iqbal, & Wan, 2023). To ensure data quality and validity, several measures aiming at identifying potentially problematic responses were implemented such as attention checks, logic validations, verification calls, response time monitoring, and IP verification. Following established research protocols, we assured all participants of their anonymity and the strictly academic use of collected data. We excluded samples of firms with incomplete data and potential errors, resulting in a valid sample consisting of 178 machine-building firms. The effective response rate of 22.3% is deemed reasonable and comparable to previous survey research of innovation in the context of China (An, Zhao, Cao, Zhang, & Liu, 2018; Tang, Nadkarni, Wei, & Zhang, 2021).

Our final sample of 178 firms comprises 124 responses collected through online surveys (69.7%) and 54 responses from onsite interviews (30.3%). We compared CEO cognitive flexibility scores between online and onsite data collection methods and the results showed no significant difference ($t = -0.084$, $p = 0.933$), confirming that the online survey design provides valid measurements of CEO cognitive flexibility comparable to onsite approach. We also assessed potential method bias by running t tests to compare online and onsite responses (Jean et al., 2021). The results show that there is no significant difference in organizational and CEO characteristics such as firm size, firm age, R&D intensity, CEO gender, CEO age, CEO tenure, and CEO education (all $p > 0.10$). Therefore, we suggested that method bias is not a concern. To further assess potential non-response bias, we compared early and late respondents following Armstrong and Overton's (1977) procedure. We divided respondents into two groups based on their response timing: those who responded within the first third of the response period versus those who responded in the last third. Independent samples t-tests were conducted on the above organizational and CEO characteristics. No significant differences were found between early and late respondents, suggesting that non-response bias was not a concern. The dataset supporting the findings of this study is available in the Open Science Framework (DOI: 10.17605/OSF.IO/9M5UG).

Measures

Our measurement scales were constructed based on earlier literature. We required respondents to truthfully answer and indicated to them that the data is confidential and completed anonymously. Seven-point Likert scales from 1 (strongly disagree) to 7 (strongly agree) were used. Table 2 presents a comprehensive overview of all the measures.

Dependent variable

Our study employs two dependent variables to comprehensively capture digital product innovation. First, we examined firms' digital patent portfolios to objectively measure innovation generation. Following recent empirical studies (Liu et al., 2021; Wang, Li, Tian, & Hou, 2023), we obtained detailed patent information for each firm from the China National Intellectual Property Administration (CNIPA). Two coders with expertise in digital product innovation independently analyzed the descriptive content of patents including technical claims and detailed descriptions, which provide information on their functions, processes, features, and technology domains. Patents were categorized as digital product innovations if they demonstrated substantial integration of digital technologies and components into physical products (Liu et al., 2021).[1] In the actual coding, the coders achieved a Krippendorf's α of 0.82, indicating strong reliability. All coding disagreements

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Table 2. Construct measures and reliability index

Constructs and measurements	Factor loading
CEO cognitive flexibility (Cronbach's $\alpha = 0.913$, CR $= 0.924$, AVE $= 0.502$)	
1. I can communicate an idea in many different ways.	0.698
2. I avoid new and unusual situations. (R)	0.731
3. I feel like I never get to make decisions. (R)	0.722
4. I can find workable solutions to seemingly unsolvable problems.	0.688
5. I seldom have choices when deciding how to behave. (R)	0.725
6. I am willing to work at creative solutions to problems.	0.747
7. In any given situation, I am able to act appropriately.	0.707
8. My behavior is a result of conscious decisions that I make.	0.692
9. I have many possible ways of behaving in any given situation.	0.700
10. I have difficulty using my knowledge on a given topic in real life situations. (R)	0.693
11. I am willing to listen and consider alternatives for handling a problem.	0.687
12. I have the self-confidence necessary to try different ways of behaving.	0.707
CEO boundary spanning (Cronbach's $\alpha = 0.849$, CR $= 0.879$, AVE $= 0.644$)	
1. I solicit information and resources from external channels.	0.820
2. I try to influence important external actors.	0.803
3. I value making use of relationships with others to support firms	0.819
4. I rely on actively solicited information and resources beyond what comes through official channels.	0.768
Firm social capital (Cronbach's $\alpha = 0.857$, CR $= 0.877$, AVE $= 0.641$)	
1. Our firm has a good reputation in the industry.	0.796
2. Our firm is well connected to other companies in the industry.	0.786
3. Our firm has access to valuable resources through the business relationships.	0.799
4. Our firm has possibility to acquire and leverage valuable information from their business relationships.	0.821
Digital product innovation performance (Cronbach's $\alpha = 0.808$, CR $= 0.801$, AVE $= 0.574$)	
1. Our digital product innovations achieve a greater success rate relative to our competitors.	0.753
2. Our digital product innovations achieve greater revenue relative to our competitors.	0.715
3. Our digital product innovations achieve greater profitability relative to our competitors.	0.802

were documented and discussed until reaching unanimous consensus. We then calculated the count of new digital patents granted to each firm within two years of our survey and lagged it by one year to ensure the temporal precedence of the predictors relative to the predicted effect (similar analyses were also conducted with applied patent data and are available upon request).

Second, to complement our patent-based measure, we also examined performance metrics, which capture how successfully these innovations are commercialized in the market. We measured digital product innovation performance using three items developed by Pesch, Endres, and Bouncken (2021). At the beginning of the survey, to ensure consistent understanding, we provided respondents with a clear definition of digital product innovation aligned with the conceptualization noted earlier (Lyytinen et al., 2016). We asked respondents to evaluate their firm's digital product innovation relative to competitors across three dimensions: success rate, revenue generation, and profitability.

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Independent variable

The independent variable in this study is CEO cognitive flexibility. This construct captures executives' capacity to adapt their decision-making approaches and attention allocation patterns in response to dynamic business environments. We measured CEO cognitive flexibility by adopting a twelve-item scale developed and validated by Martin and Rubin (1995). This scale has been extensively employed in several management research (e.g., Kiss et al., 2020; Liu et al., 2024).

Moderators

CEO boundary spanning. To access CEO boundary spanning, we used a four-item scale developed by Faraj and Yan (2009), but their study focused on boundary spanning activities at the team level. It was therefore modified to capture the current context of our research. Specifically, we conducted interviews with ten CEOs participating in the survey on the theme of boundary spanning activities to refine the wording of the measurement items and ensure that they are consistent with the CEO's actual activity scenario. Next, we made necessary modifications to the items and statement sentences taking into account the interview content, including specifying that the boundary spanning activities were initiated and implemented by the CEO.

CEO boundary spanning. Firm social capital. It reflects the firm's pool of relationship networks available to access information and resources. Drawing on existing well-established scales (Gölgeci & Kuivalainen, 2020; Zahra, 2010), we measured firm social capital using a four-item scale.

Control variables

We controlled for multiple variables that are commonly cited as influencing digital product innovation. At the CEO level, we controlled for *CEO gender* and *age*, which are directly related to their level of risk-taking and the likelihood that they are willing to invest in digital product innovation activities (Tang et al., 2021). Given that long-tenured CEOs tend to devote their efforts on information searches within the firm and engage in exploitative innovations (Kiss et al., 2020), we controlled for *CEO tenure*, measured as the number of years spent at the firm. *CEO education* is controlled as it influences their receptivity to learning about digital innovation knowledge (Firk et al., 2022). It was measured as the CEO's highest level of education, categorized into five groups: high school or below, associates, bachelors, masters, and doctorate.

At the firm level, we controlled for *firm size* (the number of employees) because large firms can leverage their pool of available resources to provide more organizational support for pursuing digital product innovation. Prior research has shown that younger firms are more flexible and less rigid (Eggers & Kaplan, 2009), which are more likely to adapt rapidly to the technological disruption than older firms (Pesch et al., 2021). We controlled for *firm age*, which is measured by the number of years since the firm's inception. *R&D intensity*, the ratio of R&D expenditure to total sales, is included since innovative firms are more inclined to pursue novel technologies and innovations. Strategic agility enables firms to achieve rapid response to adapt to new technological requirements and market demands (Del Giudice, Scuotto, Papa, Tarba, Bresciani, & Warkentin, 2021). We controlled for *strategic agility*, measured with a three-item scale developed by Ferraris et al. (2022) (Cronbach's $\alpha = 0.866$). Information technology (IT) capability, which enables firms to mobilize and deploy IT resources to cope with turbulent environments, is positively related to firms' digital product innovation outcome (Wiesböck, Hess, & Spanjol, 2020). Thus, we measured *IT capability* using a scale introduced by Lu and Ramamurthy (2011) to control for this effect (Cronbach's $\alpha = 0.862$).

Results

Common method bias

The informant-based survey data collection method has the risk of generating common method bias (CMB). Our study attempted to minimize CMB by adopting several measures, such as protecting respondents' anonymity and collecting data over different time frames (Guide & Ketokivi, 2015). Several post-hoc statistical tests were also conducted. Factor analysis was conducted on all items using Harman's single-factor test method. It was observed that the total interpreted variation of the samples was 63.106%, and the first principal component without rotation contributed to 30.007%, which was deemed acceptable. To further validate this possibility, we included a common latent factor (CLF) in the measurement model to compare the fit of the model with and without the CLF. The results indicated that the model fit indices after adding the CLF (χ^2/df = 1.260, IFI = 0.969, TLI = 0.965, CFI = 0.969; RMSEA = 0.038) were not significantly better than that without the CLF (χ^2/df = 1.390, IFI = 0.954, TLI = 0.947, CFI = 0.953; RMSEA = 0.047). The small differences (<0.200) (Han & Zhang, 2021) suggested that our results remained unaffected by CMB.

Assessment of measurement model

We conducted a test to identify the reliability, convergent validity, and discriminant validity of the measures. As shown in Table 2, the Cronbach's α coefficients for each measure were within the acceptable range, showing satisfactory reliability. All constructs exhibit a composite reliability value exceeding 0.70, indicating robust internal consistency. Regarding the convergent validity, the factor loading of each item and the average variance extraction (AVE) values were greater than the cut-off point of 0.50. It provides support to the assertion that the indicator is applicable to all structures. Furthermore, all pairs of constructs were subjected to Fornell and Larcker's (1981) discriminant validity test. The findings reveal that the AVE's square roots for each construct exceed the inter-scale correlations (see Table 3), providing strong evidence for discriminant validity of our core constructs.

Hypotheses test

Table 3 reports the descriptive statistics and correlations for all variables in our model. We mean-centered the independent variable and moderators before creating interactions to avoid the multicollinearity issue. We used hierarchical regression analyses to test the hypotheses. For each dependent variable, we first reported the null models that only include control variables and the main effect models that then including the independent variable. We then added and examined the interaction of CEO cognitive flexibility with CEO boundary spanning and firm social capital, respectively. Finally, we also reported the full model containing all interactions in this study, which serves as the basis for hypothesis testing given consistent results across models. The variance inflation factors (VIF) for all study variables were within a reasonable range, with an average of 1.21 and a high of 1.37, well below the acceptable level of 10. Therefore, multicollinearity is not a serious concern.[2]

Table 4 displays the results of the hierarchical regression analyses. For digital patent volume, CEO cognitive flexibility showed a positive relationship (Model 5: $\beta = 0.836$, $p < 0.01$). Similarly, for digital product innovation performance, CEO cognitive flexibility demonstrated a positive effect (Model 10: $\beta = 0.309$, $p < 0.01$). These consistent findings across both dependent variables provide robust support for H1.

Regarding the predicted moderations, the results show that both CEO boundary spanning (Model 5: $\beta = 0.393$, $p < 0.01$ for patent; Model 10: $\beta = 0.121$, $p < 0.05$ for performance) and firm social capital (Model 5: $\beta = 0.375$, $p < 0.05$ for patent; Model 10: $\beta = 0.186$, $p < 0.01$ for performance) strengthen the relationship between CEO cognitive flexibility and both measures of digital product innovation. Thus, H2 and H3 are supported. Additionally, to provide a more nuanced understanding of the moderating effects, we employed the Johnson-Neyman (J-N) technique to identify the regions of significance (Preacher, Curran, & Bauer, 2006). The relationship between CEO cognitive flexibility and digital patent volume becomes statistically significant when the confidence interval (the colored part in Figs. 2 and 3) does not include zero (similar analyses were also conducted with digital product innovation performance as the dependent variable and are available upon request). As Figure 2 reveals, the simple slope of CEO cognitive flexibility on digital patent volume becomes significant when the mean-centered CEO boundary spanning value is either below −3.12 or exceeds −0.59.

Table 3. Descriptive statistics and correlation matrix

	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1. Firm size (log)	–													
2. Firm age (log)	0.09	–												
3. R&D intensity	−0.01	0.08	–											
4. CEO gender	0.00	0.00	0.06	–										
5. CEO age	0.22*	−0.04	−0.02	0.09	–									
6. CEO tenure	−0.01	0.20*	−0.05	0.06	−0.09	–								
7. CEO education	0.41*	−0.05	−0.11	−0.03	0.25*	−0.05	–							
8. Strategic agility	−0.01	−0.04	0.11	−0.02	0.10	−0.07	0.05	(0.887)						
9. IT capability	0.10	−0.05	−0.09	−0.01	0.06	0.05	0.05	0.16*	(0.883)					
10. CEO cognitive flexibility	0.16*	0.07	0.09	0.05	0.21*	0.07	0.08	0.02	0.05	(0.709)				
11. CEO boundary spanning	0.25*	−0.12	0.05	0.11	0.17*	−0.13	0.31*	0.10	0.04	0.14	(0.802)			
12. Firm social capital	0.14	−0.14	0.07	0.17*	0.09	−0.04	0.18*	0.12	0.23*	0.08	0.38*	(0.801)		
13. Digital product innovation performance	0.30*	−0.03	0.06	0.00	0.40*	−0.14	0.31*	0.19*	0.19*	0.36*	0.37*	0.48*	(0.758)	
14. Digital patent volume	0.13	−0.13	0.01	−0.01	0.21*	0.03	0.22*	0.07	0.08	0.36*	0.32*	0.32*	0.58*	–
Mean	6.26	3.15	2.91	0.80	49.17	6.03	3.42	3.94	3.79	4.21	4.42	3.96	3.61	3.85
SD	0.83	0.26	2.02	0.40	7.08	2.76	0.87	1.29	1.29	1.10	1.43	1.47	1.55	3.36

Notes: $N = 178$. *$p < 0.05$ (two-tailed). Bracketed values on the diagonal are the square root of the AVE value of each scale.

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Table 4. Hierarchical regression results

	Digital patent volume					Digital product innovation performance				
	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10
Firm size (log)	0.173	0.027	−0.248	−0.269	−0.354	0.287**	0.233*	0.118	0.087	0.065
	(0.331)	(0.314)	(0.297)	(0.297)	(0.292)	(0.136)	(0.131)	(0.123)	(0.113)	(0.112)
Firm age (log)	−1.810*	−2.020**	−1.382	−1.218	−1.048	−0.011	−0.089	0.178	0.371	0.416
	(0.972)	(0.920)	(0.865)	(0.865)	(0.847)	(0.400)	(0.382)	(0.359)	(0.328)	(0.326)
R&D intensity	0.079	0.027	−0.016	−0.013	−0.030	0.065	0.046	0.028	0.017	0.012
	(0.125)	(0.119)	(0.111)	(0.110)	(0.108)	(0.051)	(0.049)	(0.046)	(0.042)	(0.042)
CEO gender	−0.180	−0.250	−0.469	−0.563	−0.613	−0.062	−0.088	−0.182	−0.321	−0.331
	(0.616)	(0.582)	(0.545)	(0.548)	(0.537)	(0.253)	(0.242)	(0.227)	(0.208)	(0.206)
CEO age	0.075**	0.044	0.064*	0.056*	0.066**	0.065***	0.053***	0.062***	0.059***	0.062***
	(0.037)	(0.035)	(0.033)	(0.033)	(0.032)	(0.015)	(0.015)	(0.014)	(0.012)	(0.012)
CEO tenure	0.098	0.064	0.084	0.088	0.093	−0.055	−0.067*	−0.058*	−0.055*	−0.055*
	(0.092)	(0.087)	(0.081)	(0.081)	(0.079)	(0.038)	(0.036)	(0.034)	(0.031)	(0.031)
CEO education	0.606*	0.602**	0.495*	0.483*	0.454	0.294**	0.292**	0.245**	0.200*	0.199*
	(0.317)	(0.300)	(0.287)	(0.280)	(0.280)	(0.130)	(0.125)	(0.119)	(0.107)	(0.108)
Strategic agility	0.095	0.102	−0.045	−0.031	−0.086	0.143*	0.145*	0.085	0.076	0.060
	(0.196)	(0.185)	(0.174)	(0.173)	(0.170)	(0.080)	(0.077)	(0.072)	(0.066)	(0.065)
IT capability	0.130	0.100	0.166	0.056	0.109	0.167**	0.156**	0.183**	0.098	0.113*
	(0.196)	(0.185)	(0.173)	(0.178)	(0.175)	(0.081)	(0.077)	(0.072)	(0.068)	(0.067)
CEO cognitive flexibility (H1)		1.015***	0.805***	0.956***	0.836***		0.377***	0.290***	0.344***	0.309***
		(0.221)	(0.209)	(0.205)	(0.204)		(0.092)	(0.087)	(0.078)	(0.078)
CEO boundary spanning			0.430**		0.264			0.186***		0.057
			(0.168)		(0.172)			(0.070)		(0.066)
CEO cognitive flexibility **X** CEO boundary spanning (H2)			0.580***		0.393***			0.237***		0.121**
			(0.139)		(0.147)			(0.058)		(0.057)

(*Continued*)

Table 4. (*Continued.*)

	Digital patent volume					Digital product innovation performance				
	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10
Firm social capital				0.529***	0.380**				0.389***	0.350***
				(0.159)	(0.165)				(0.061)	(0.063)
CEO cognitive flexibility				0.560***	0.375**				0.239***	0.186***
X Firm social capital (H3)				(0.143)	(0.152)				(0.054)	(0.059)
Constant	1.184	0.522	−1.063	−1.366	−1.824	−3.342**	−3.588**	−4.259***	−4.837***	−4.944***
	(3.834)	(3.626)	(3.394)	(3.385)	(3.316)	(1.576)	(1.507)	(1.411)	(1.286)	(1.276)
Observations	178	178	178	178	178	178	178	178	178	178
Adjusted R^2	0.053	0.154	0.268	0.275	0.308	0.247	0.312	0.405	0.507	0.517

Notes: *$p < 0.10$, **$p < 0.05$, ***$p < 0.01$ (two-tailed tests). Standard errors in parentheses.

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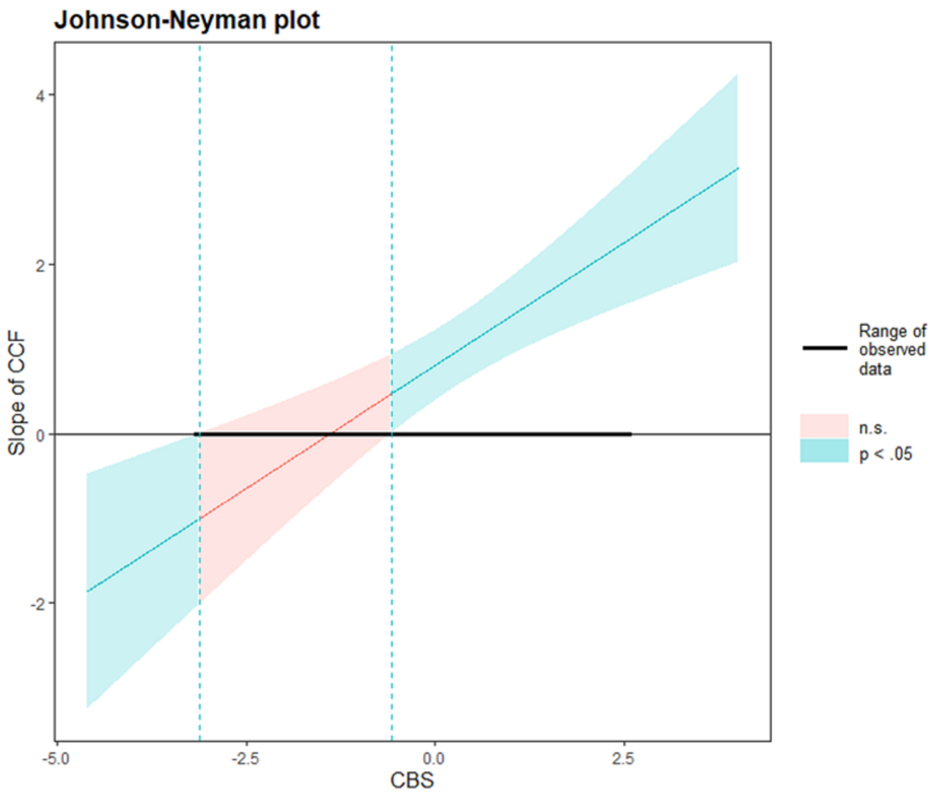

Figure 2. The moderating effect of CEO boundary spanning
Notes: CCF, CBS represent CEO cognitive flexibility, CEO boundary spanning after centralization, respectively.

Furthermore, the magnitude of this positive relationship strengthens with higher levels of boundary spanning, indicating an amplifying moderation pattern. Similarly, the J-N analysis for firm social capital (Fig. 3) showed that the simple slope of CEO cognitive flexibility on digital patent volume becomes significant when firm social capital is exceeds −0.86, with the effect magnitude intensifying at higher levels of social capital. These findings confirm the positive moderating effect of CEO boundary spanning and firm social capital.

Study 2: Experimental Research

While Study 1 demonstrated that cognitively flexible CEOs improve their firms' digital product innovation outcomes, the survey measures and data do not allow us to explicitly establish causality and assess the invisible decision-making processes of firm executives (Song, Cadsby, & Bi, 2012). To address these limitations and complement our field survey findings, Study 2 focuses specifically on the fundamental cognitive mechanism that initiates this process: how cognitive flexibility enhances the acquisition and processing of insightful information. This information processing capability represents the foundational mechanism through which cognitively flexible CEOs recognize opportunities and make decisions to pursue digitization, enabling the innovation generation and commercialization outcomes. Specifically, we measured cognitive flexibility through experimental tasks rather than survey items, and simulated the psychological conditions individuals face when making decisions about digital product innovation trajectories in incumbent firms. This experimental design enables

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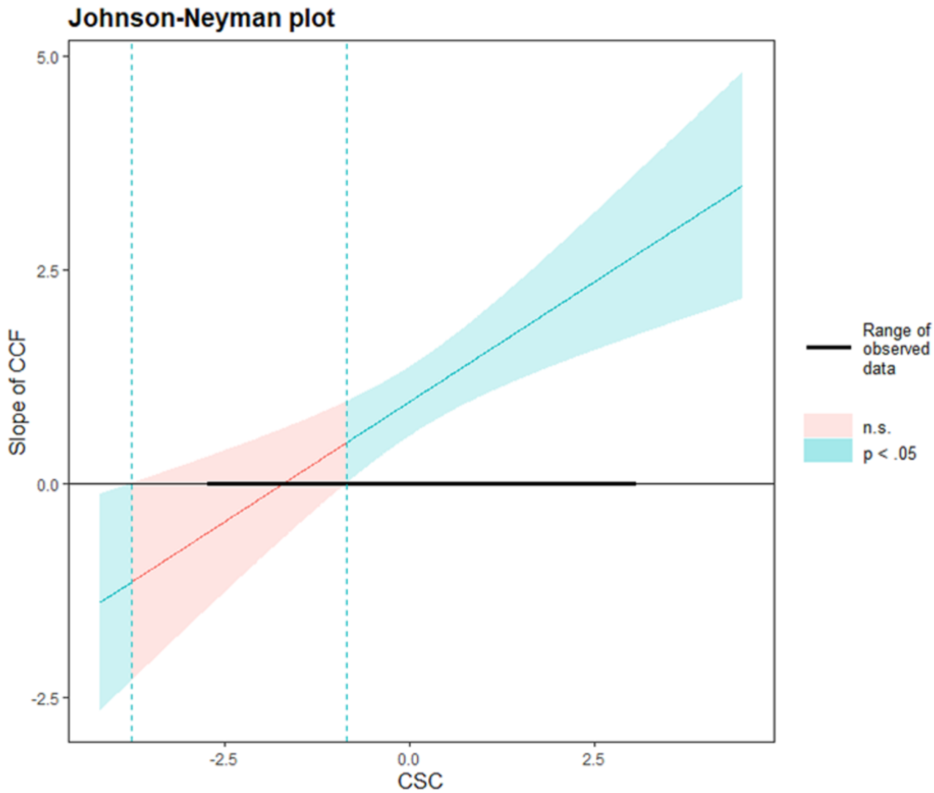

Figure 3. The moderating effect of firm social capital
Notes: CCF, CSC represent CEO cognitive flexibility, firm social capital after centralization, respectively.

us to test the mediating role of insightful information acquisition in linking cognitive flexibility to digital product innovation, while establishing causality through controlled manipulation.

Sample

We recruited 134 undergraduate students to comprise our sample. The selection of this sample was based on several criteria. First, according to the effect sizes from pilot tests, power analyses indicated that a sample of 128 usable respondents was necessary. Second, undergraduate students typically lack decision-making experience in relevant market environments and are more likely to respond based on the mechanisms provided by the experimental setting, making the manipulation less difficult. They have a relatively steep learning curve and low likelihood of opportunistic behavior than professional managers (e.g., MBA students). We recruited 150 undergraduate students from a major university in China and offered them monetary rewards for completing the experimental tasks. Sixteen participants failed the attention check, yielding a final sample of 134 participants ($M_{\text{age}} = 20.66$, $SD_{\text{age}} = 1.72$). Approximately 57% are female.

Procedure

Participants were invited to engage in a 'Digital Product Innovation Decision-Making Exercise' in which they were asked to assume the role of CEO of the 'Refrigerator Manufacturing Company', a firm that produces traditional refrigerators. We explained the job description and the product characteristics of the refrigerator to the participants in detail. We administered a number-letter switching

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task to all participants (see Appendix A), which is a commonly used experimental task to measure the level of cognitive flexibility (Rogers & Monsell, 1995). Based on the swift cost of reaction time, we distinguished between two groups of high and low cognitive flexibility. They then needed to complete a purportedly irrelevant survey that consisted of measures of insightful information acquisition. Next, participants were requested to review a fictitious segment from a research report centered on the topic of 'digital product innovation' released by a leading consulting firm. This research report underscores the significance of digital product innovation as a leading technology trend, while noting the substantial difference between digital product innovation and traditional innovation, necessitating firms to adjust accordingly (see Appendix B). The original version of the research report is in Chinese, and the appendix shows our translated English version. Two doctoral candidates did the translation simultaneously, and discussed the inconsistent statements until a consensus was reached. Finally, they completed a simple questionnaire that includes demographic inquiries and a survey pertaining to decisions on digital product innovation.

Measures

Digital product innovation

Building on previous research (King & Kugler, 1993; Teo, Wei, & Benbasat, 2003) and the specific experimental scenario of this study, we measured digital product innovation using a 3-item scale that captures participants' evaluation and pursuit of digital product innovation. One sample item was 'I would like to invest in numerous resources to develop digital product innovation' (1 'strongly disagree' to 7 'strongly agree'; Cronbach's $\alpha = 0.74$).

Insightful information acquisition

Insightful information is partly a reflection of the quality of information available to decision-makers, which can be analyzed from multiple perspectives to gain unconventional and novel insights. Drawing on Maltz's (2000) approach, we adapted a 4-item scale to measure insightful information acquisition. One sample item was 'I often acquire information that can be interpreted from many perspectives' (1 'strongly disagree' to 7 'strongly agree'; Cronbach's $\alpha = 0.81$).

Control variables

We controlled for several factors that could influence participants to make innovation decisions, including age, age squared, and gender.

Results

The descriptive statistics and correlations for Study 2 variables are presented in Table 5. First, we examined whether cognitive flexibility influences digital product innovation. Consistent with our theoretical predictions and Study 1 findings, participants with high cognitive flexibility ($M = 4.82$, $SD = 1.38$) were more inclined to pursue digital product innovation relative to those with low cognitive flexibility ($M = 3.93$, $SD = 1.68$; $t(132) = 28.47$, $p = 0.001$). Moreover, with the inclusion of control variables, the results remained robust ($B = 0.65$, $SE = 0.24$, $p < 0.01$).

To test the potential mediating role of insightful information acquisition, we used model 4 in the PROCESS macro (Hayes, 2013). We conducted a bootstrap analysis using 5,000 random samples and interpreted the results based on the 95% confidence intervals (CIs). Figure 4 displays the path coefficients, indirect effect, total effect, and CIs. We conducted testing on this model with and without control variables, and found consistent results. The model with control variables is depicted in Figure 4. The indirect effect between cognitive flexibility and digital product innovation through insightful information acquisition was significant ($B = 0.37$, $SE = 0.15$, CI = [0.09, 0.68]). As such, the mediating effect of insightful information acquisition was supported.

Table 5. Means, standard deviations, and correlations, supplementary study

Variable	M	SD	1	2	3	4	5
1. Cognitive flexibility	0.50	0.50	–				
2. Digital product innovation	4.37	1.59	0.28*	(0.81)			
3. Insightful information acquisition	4.22	1.70	0.23*	0.55*	(0.80)		
4. Age	20.66	1.72	0.04	−0.01	0.11	–	
5. Gender	1.52	0.50	−0.33*	−0.03	−0.07	0.04	–

Notes: $N = 134$; *$p < 0.01$ (two-tailed). Bracketed values on the diagonal are the square root of the AVE value of each scale.

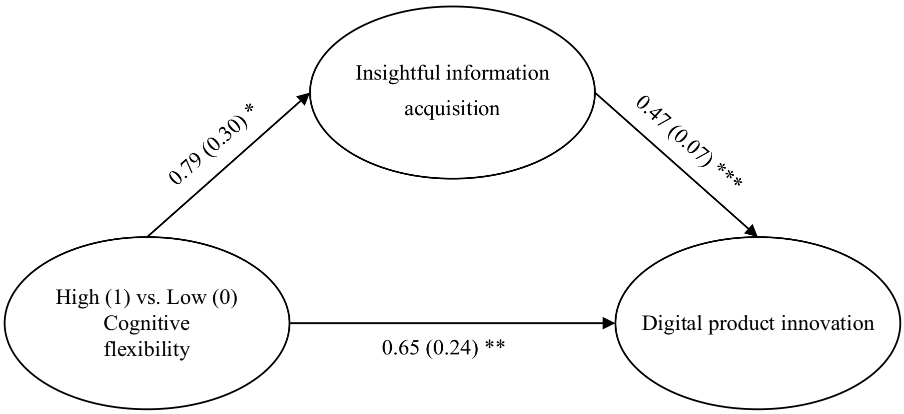

Figure 4. Mediation analysis, supplementary study
Notes: Indirect effect: 0.37 (0.15), CI [0.09, 0.68]; total effect: 1.02 (0.28), CI [0.47, 1.56]. * $P < 0.05$ ** $p < 0.01$ *** $p < 0.001$.

Discussion

An increasingly digitized world poses new challenges on how to organize innovation, substantially transforming traditional product development paradigms and demanding novel approaches to value creation (Verganti et al., 2020). For incumbent firms, this transformation manifests prominently in the realm of digital product innovation, where they need to navigate the transition from established manufacturing practices to fluid, iterative development processes (Volberda et al., 2021). Yet their executives typically rely on fixed cognitive representations deeply ingrained in historical manufacturing contexts, leading to struggles in dealing with the distinctive demands in digital product innovation. Drawing on upper echelons theory and the ABV, this study conducts an empirical inquiry into the role of cognition, and more specifically the role of cognitive flexibility in enabling firms to pursue digital product innovation. Through a mixed-methods approach combining field survey data and experimental evidence, we find that cognitive flexibility enables CEOs to meet the distinctive demands of digital product innovation by facilitating insightful information acquisition and processing. This relationship is strengthened when supported by boundary spanning activities and firm social capital, which serve as critical attention structures that shape how CEOs access and interpret information. These findings offer several important theoretical and practical implications.

Theoretical Implications

Our research makes important contributions to several areas of academic studies. First, we contribute to digital innovation literature by advancing understanding of the cognitive microfoundations that

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enable incumbent firms to successfully pursue digital product innovation. While previous research has primarily focused on organizational-level factors such as structure, business models, and alliance networks (e.g., Bockelmann et al., 2024; Ceipek et al., 2021), scholars have increasingly called for investigation into the role of individual actors in digital product innovation processes (Bunduchi et al., 2022; Kohli & Melville, 2019). Our study responds to these calls by illuminating how executive cognitive characteristics – specifically cognitive flexibility – influence firms' capacity to dynamically manage the distinctive demands of digital product innovation. Our findings reveal how cognitive flexibility serves as a crucial mechanism enabling CEOs to reconcile seemingly contradictory demands inherent in digital product innovation: balancing fixed manufacturing constraints with fluid digital architectures, integrating product-centric with ecosystem-oriented thinking, and managing both predetermined and emergent value creation logics. This helps explain why some incumbent firms successfully adapt to digital product innovation while others remain trapped in traditional development approaches despite similar organizational capabilities and resources.

Second, we contribute to the ABV by elucidating how cognitive flexibility shapes attention allocation in complex digital environments. While the ABV has traditionally emphasized how organizational structures influence attention patterns (Joseph & Wilson, 2018; Ocasio, 1997, 2011), our research extends this perspective by demonstrating the dynamic interplay between individual cognitive capabilities and structural elements in determining innovation outcomes. Specifically, we show that cognitive flexibility works through two distinct attention structures identified in ABV: communication channels (manifested in boundary spanning) and social relationships (embodied in firm social capital). This advances recent theoretical developments on the situated nature of attention (Brielmaier & Friesl, 2023; Ocasio et al., 2018) by demonstrating how cognitive flexibility enables executives to effectively adjust their attentional focus between traditional product development cycles and the fluid evolution of digital products, while simultaneously leveraging structural mechanisms to access and interpret diverse information streams. This contribution enhances our understanding of how attention processes operate at the intersection of individual cognition and organizational structures, particularly in contexts requiring balance between traditional and digital product innovation logics.

Thirdly, we contribute to the strategic leadership research on the role of individual cognition in organizational adaptation (Eggers & Kaplan, 2013; Hambrick, 2007) by how cognitive flexibility shapes strategic decision-making processes in the face of evolving digital contexts. Specifically, we demonstrate that cognitive flexibility enables CEOs to extract novel insights and interpret information from multiple perspectives, which helps challenge existing assumptions and ultimately pursue digital product innovation. Prior research has emphasized the importance of understanding the cognitive underpinnings of organizational adaptation (Eggers & Kaplan, 2009; Tripsas & Gavetti, 2000; Vuori & Huy, 2016), particularly in contexts requiring significant departures from established practices. Our study thus advances this literature by theoretically and empirically demonstrating that cognitive flexibility operates as a crucial yet underexplored adaptive cognitive capability that alters how executives dynamically reconfigure their decision-making approaches in response to novel digitization demands (Kiss et al., 2020; Laureiro-Martínez & Brusoni, 2018). In doing so, we provide actionable insights into which cognitive characteristics matter for navigating digitization, moving beyond general calls for 'cognitive adaptability' to specify a measurable construct (Firk et al., 2022; Volberda et al., 2021).

Practical Implications

Our findings offer several practical insights for incumbent firms pursuing digital product innovation. First, we suggest that cognitive flexibility should be an important consideration in executive selection and development, particularly for firms seeking to pursue digital product innovation. Organizations might benefit from assessing cognitive flexibility as part of their executive recruitment processes and developing training programs to enhance this capability among existing leaders. Second, our results highlight the importance of creating organizational conditions that amplify the benefits of cognitive flexibility. Firms should encourage CEO boundary spanning activities by creating formal mechanisms for external engagement and knowledge acquisition. Similarly, investments in building and maintaining social capital through strong stakeholder relationships can enhance the firm's capacity to leverage its CEO's cognitive flexibility for digital product innovation. Third, our findings suggest that incumbent firms might benefit from recognizing and addressing the cognitive demands of digital product innovation. This could involve developing organizational practices that support flexible thinking and rapid adaptation, such as experimental learning approaches, cross-functional collaboration, and iterative development processes that align with digital product innovation's fluid and iterative nature.

Limitations and Future Research Directions

This study offers several limitations and opportunities for future research. First, while our study illuminates the role of cognitive flexibility in digital product innovation, we have not addressed the broader array of cognitive characteristics. In particular, executives in incumbent firms often carry deeply ingrained beliefs and decision-making patterns developed through years of success with traditional product development approaches (Firk et al., 2022). These cognitive biases, such as overconfidence in existing capabilities or anchoring to past successful strategies, may significantly influence digital product innovation processes and outcomes independently of cognitive flexibility. Future research could explore how various cognitive biases shape incumbent firms' digital product innovation trajectories.

Second, although we investigate firm social capital as an organizational-level attention mechanism, we do not address how individual-level social relationships, particularly CEOs' personal networks, might shape attention allocation in digital product innovation. CEOs' personal ties could serve as crucial channels for accessing and filtering information about digital product innovation (Maula et al., 2013). These individual networks likely differ from organizational social capital in both their nature and influence on attention patterns. Future research could explore how CEOs' personal social networks function as attention structures in digital contexts.

Third, while we employed complementary measures to strengthen construct validity through methodological triangulation, this approach also presents limitations. The self-report measure in Study 1 (Martin & Rubin, 1995), though ecologically valid for capturing CEOs' behavioral tendencies in organizational contexts, may be subject to social desirability bias. Conversely, the experimental task-switching measure in Study 2 (Rogers & Monsell, 1995), while providing assessment of cognitive switching ability under controlled conditions, may not completely reflect the contextual and motivational factors in actual organizational decision-making (Laureiro-Martínez & Brusoni, 2018). These measurement differences, though theoretically justified by our research designs, mean that we cannot claim that both studies capture the identical aspects of cognitive flexibility. Future research could benefit from employing multiple measures of cognitive flexibility within a single study design.

Finally, our contextual focus on machine-building firms in China may constrain the generalizability of our findings. However, several factors suggest this limitation may not undermine our core findings. First, our random sampling across three provinces with different manufacturing development levels captures meaningful heterogeneity, providing sufficient variance for hypothesis testing. Second, our multi-method triangulation strengthens confidence beyond what sample size alone could provide; convergent findings across perceptual and patent measures in Study 1, combined with causal evidence from Study 2, offer robust support for our hypotheses. Third, consistency between our online and on-site data collection methods, along with non-response bias tests showing no systematic differences, suggests our findings reflect stable patterns rather than sampling artifacts. Moreover, our focus on machine-building firms offers a theoretically relevant context where tensions between traditional manufacturing and digital product innovation are pronounced, making it an ideal setting. Future research could extend our findings to larger, multi-industry samples and different organizational contexts to establish broader generalizability.

Data availability statement. In line with the current standards of methodological transparency, we have made the data, syntax, and results available at https://osf.io/9m5ug/

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Notes

1. Our coding protocol identified digital product innovations as patents demonstrating: (1) embedded processors, AI, or software enabling autonomous or adaptive product behavior; (2) IoT, cloud, or network features enabling data exchange and remote functionality; (3) sensors combined with analytics that generate insights or enable new services and so on.
2. We also estimated all models including industry dummy variables to account for potential industry-specific effects within the machine-building sector. The results remained consistent with our main findings.

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Appendix A. Measures of Cognitive Flexibility: Number-Letter Switching Tasks

Cognitive flexibility reflects people's ability to adjust their cognitive resources to adjust their behavioral responses in response to changes in the external environment. Experimental tasks frequently used in previous studies to measure the level of cognitive flexibility include the Wisconsin Card Sorting Task (WCST) and the switching task (Deák & Wiseheart, 2015). However, recent studies have indicated that assessing cognitive flexibility through the switching task provides a more refined measurement. This measurement paradigm is commonly consistent across studies, offering easier interpretation and comprehension (Lange, Kip, Klein, Müller, Seer, & Kopp, 2018). Therefore, this study takes the switching cost in the switching task as the basis to distinguish subjects with different levels of cognitive flexibility, and the 'number-letter switching task' of Rogers and Monsell (1995) as the specific implementation task.

Specifically, a square grid appears in the center of the screen. Each stimulus is composed of a letter and a number (such as 4E or M5), and the sequence of numbers and letters is random (as shown in Fig. A1). Letters may be vowels (A/E/I/U) or consonants (G/K/M/R); numbers may be odd (3/5/7/9) or even (2/4/6/8), with random combinations of letters and numbers. The task includes an exercise and a formal experiment. The exercise consists of three parts: a letter-only judgment task, a number-only judgment task, and a joint number-letter task, with the order of appearance of the letter-only and number-only judgment tasks counterbalanced between subjects. A total of 32 trials are included in the letter-only judgment exercise, i.e., 16 trials for consonants (half paired with odd numbers and half paired with even numbers) and 16 trials for vowels (half paired with odd numbers and half paired with even numbers). The number-letter stimulus pairs always appear in the upper two squares, and subjects are asked to press the E or I key as quickly and accurately as possible to determine whether the letter is a consonant or a vowel. An '×' appears after an incorrect response and prompts for the correct key to be pressed, and the subject must re-press the key to correct the incorrect response. The number-only judgment exercise is similar to the letter-only exercise, except that the number-letter stimulus pairs always appear in the lower two squares, and subjects are asked to press the E or I key as quickly and accurately as possible to determine whether the number is even or odd. In the joint number-letter exercise task, stimulus pairs are presented one by one in a clockwise direction, and the next number or letter is not the same as the previous one. Subjects are informed before the start of the experiment to do a letter judgment task (i.e., press E or I to determine whether a letter is a consonant or a vowel) when the stimulus pairs appear in the upper two squares, and to do a number judgment task (i.e., press E or I to determine whether a number is even or odd) when the stimulus pairs appear in the lower two squares. A minimum of 80% correct on the joint exercise task was required to enter the formal experiment.

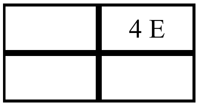

Figure A1. Example of stimulus presentation for a number-letter task

The formal experiment requires the completion of 128 joint numeric-letter tasks, where an '×' appears after an incorrect keystroke, but no longer prompts for the correct key. The task switches from a numeric task to a letter task when the stimulus pair moves from the first to the second quadrant, and these two cases are referred to as switching trials. When the stimulus is from the fourth quadrant to the third quadrant, or from the second quadrant to the first quadrant, it does not need to switch the task type, which is a non-switching trial. Record the correct rate and time of the reaction, and reaction time switching cost = average reaction time for correct reactions in switching trials – average reaction time for correct reactions in non-switching trials. Next, the switching cost of reaction time is calculated, which is used as the basis for subsequent experiments to distinguish between subjects with high cognitive flexibility and low flexibility, that is, the switching cost of subjects with high cognitive flexibility is small, and that of subjects with low cognitive flexibility is large.

Appendix B. Digital Product Innovation Decision-Making Exercise

You are the chief executive officer at Refrigerator Manufacturing Company. The scope of your responsibilities encompasses various critical facets aimed at ensuring the seamless functioning of the organization, strengthening market viability and sustaining profitability. This includes leading activities related to product, technology, and sales planning; determining the direction of the company's technology innovation and product development; and overseeing the execution of the product development and innovation trajectory.

About your firm: Refrigerator Manufacturing Company's current business focuses mainly on the R&D, design, manufacture and sales of traditional refrigerators, and is committed to providing domestic and neighboring markets with affordable and stable domestic and commercial refrigerator products. Having been immersed in refrigerator manufacturing for over ten years, with the core concept of 'Craftsmanship Manufacturing, Quality Life', the company has earned a good reputation in the market through refined production management, strict quality control, and closeness to consumer demands.

Strategic tasks: Your company has been engaged in refrigerator manufacturing for many years, with relatively mature production technology and accumulated a certain number of customers. According to the company's overall strategy, your task is to continue to expand the company's market share in the field of refrigerator manufacturing.

Experimental scenario: The following is a segment from a research report released this week by a leading consulting firm. You should spend at least a one minute reading this report.

Title: Digital product innovation: Opportunities and challenges for enterprise transformation

Content: In the current era of rapid development of digital technology, digital products undoubtedly represent the future trend of product development. Digital product innovation enables firms to better understand and predict customer needs, and meet expectations through customized and personalized products, which plays a vital role in attracting customers, especially young customers. They are highly receptive to new things and have a strong desire for technology and innovative products, which provides a good market prospect for digital product innovation. Digital product innovation cannot only improve the performance and functions of products but also create a new user experience to meet the increasingly diverse needs. This is conducive to firms to stand out in the fierce product competition and occupy a favorable market position.

However, digital product innovation differs significantly from traditional approaches to product innovation. It requires firms to not only make large-scale adjustments at the technical level, but also conduct in-depth exploration in organizational practice, such as business models, user experience, data analysis. Such innovations often necessitate structural changes in firms, including organizational restructuring, reshaping of firm culture, and rapid adaptation and application of emerging technologies. Many firms accustomed to producing traditional products are encountering challenges in this transition. Roughly half of conventional firms aiming to leverage the potential of digital product innovation encounter challenges in realizing anticipated returns or even suffer losses. This is often attributed to a limited grasp of new technologies, an inability to swiftly adapt to change, and a lack of adaptability in managing innovation.

Questionnaire: Please complete a short survey combining the above report and the current state of the company's operations (1 'strongly disagree' to 7 'strongly agree'):

- I believe that digital product innovation has great potential for application in our firm.
- I believe that digital product innovation can be detrimental to our firm's current business. (R)
- I would like to invest in numerous resources to develop digital product innovation.

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